

Presenter's Guide

Cointegration & Error-Correction Dynamics in a Commodity-Currency Basket

A VECM Approach to Relative-Value Currency Allocation

Companion to `fx_cointegration_vecm.ipynb`. This document explains the project end-to-end, fully specifies every statistical test (hypotheses, statistic, distribution, decision rule), gives the relevant mathematics, and lists the headline results so you can present and defend each step.

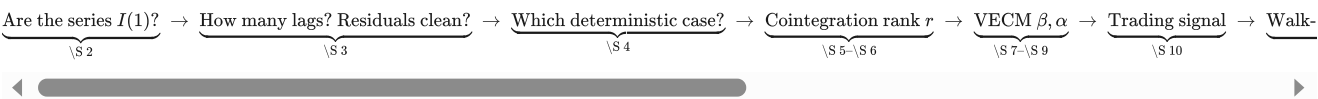
0. The 30-second pitch

We ask whether a basket of seven commodity / risk-sensitive currencies shares a **stable long-run equilibrium** (cointegration), and what the **adjustment dynamics** say about which currencies to hold. We build the **full econometric pipeline** — unit-root testing → lag selection → Johansen cointegration → VECM → out-of-sample walk-forward — and treat the *instability* of FX cointegration as the object of study rather than something to hide. The headline finding is honest: **cointegration in this basket is regime-dependent** — it exists in only ~15% of rolling windows — and while the equilibrium error shows statistically significant mean reversion, it **does not beat a random walk** out-of-sample. The grade lives in the rigour of the tests, not the size of the Sharpe.

One-sentence thesis to repeat: *the statistics are the deliverable; a negative or unstable result, properly tested, is a complete result.*

1. The narrative arc (how the sections connect)

The analysis is a **funnel** — each test earns the right to run the next:



Crucial framing point to make early. The notebook does two kinds of work with two different correctness standards:

Regime	Sections	Claim type	Standard
In-sample structural inference	§2–§9	<i>descriptive</i> ("were these cointegrated over 2016–26?")	full sample is legitimate — no forecast is made
Out-of-sample evaluation	§11–§13	<i>predictive</i> ("could you have traded it?")	strictly causal — any quantity used at time t uses only data $\leq t$

This distinction pre-empts the "isn't that look-ahead bias?" question.

2. The data

- **Source:** Bloomberg terminal pull (`src/fx_data_pull.py`), daily `PX_LAST` , ~2,600 trading days, 2016–2026.
- **Basket (commodity / risk bloc):** AUD, NZD, CAD, NOK, SEK, ZAR, MXN.
- **Transform:** every series is quoted as **USD per 1 unit of the currency** (USD-quoted pairs inverted), then **logged**. We model $y_t = \log(\text{price})$.

Why these choices matter (say this): - *Common numéraire + logs* make the log-levels directly comparable, so a cross-currency cointegrating vector $\beta'y_t$ is meaningful and its weights are interpretable as elasticities. - *Thematic tightness is statistical, not cosmetic*. A grab-bag basket tends to produce "cointegration" that is just **shared USD exposure** (the common-dollar-factor trap, tested in §8). Seven series also keeps Johansen's test powerful — it degrades badly past ~10–12 series. - *Survivorship*: the basket was chosen **ex-ante on economic theme**, not by screening for what cointegrates, so we are not data-snooping the basket.

3. Foundational mathematics (define these once)

Integration order. A series y_t is **integrated of order d** , written $y_t \sim I(d)$, if it must be differenced d times to become stationary. A unit-root (random-walk) series $y_t = y_{t-1} + \varepsilon_t$ is $I(1)$: non-stationary, but its first difference $\Delta y_t = \varepsilon_t$ is stationary $I(0)$.

Cointegration (Engle–Granger 1987). A vector of $I(1)$ series $y_t \in \mathbb{R}^n$ is **cointegrated** if there exists a vector $\beta \neq 0$ such that the linear combination

$$z_t = \beta'y_t \sim I(0)$$

is **stationary**. Economically: the series individually wander (non-stationary), but a particular weighted combination is tied together — a **long-run equilibrium** — and deviations from it are temporary. The number of independent such vectors is the **cointegration rank r** .

Why we care: if z_t is stationary it is mean-reverting, so a stretched z_t predicts a partial reversion → a tradeable "which currency is cheap/rich" signal.

4. §2 — Order-of-integration tests (is each series $I(1)$?)

Why first: cointegration is only defined for series integrated of the **same** order, conventionally $I(1)$. If a series were $I(0)$ it cannot belong to a cointegrating relation; if $I(2)$ the standard Johansen setup is invalid. So we must pin the integration order before anything else.

We test each **log level** and each **first difference** with a battery. Use a *confirmatory pair* (opposite nulls) plus power- and break-robust variants.

4.1 Augmented Dickey–Fuller (ADF) — the workhorse

Regression:

$$\Delta y_t = \mu + \delta t + \rho y_{t-1} + \sum_{i=1}^p \gamma_i \Delta y_{t-i} + \varepsilon_t$$

The lagged differences "augment" the regression to whiten ε_t .

- $H_0: \rho = 0$ — a **unit root** is present (y_t is non-stationary, $I(1)$).
- $H_1: \rho < 0$ — y_t is (trend-)stationary, $I(0)$.
- **Statistic:** $ADF = \hat{\rho}/se(\hat{\rho})$ — a t -ratio, but it does **not** follow a t -distribution under H_0 ; it follows the non-standard **Dickey–Fuller distribution** (left-tailed, more negative critical values).
- **Decision:** reject H_0 (conclude stationary) if ADF is **more negative** than the critical value / if $p < 0.05$.

4.2 KPSS — the complementary test (opposite null)

Based on the partial-sum process of residuals $S_t = \sum_{i=1}^t \hat{u}_i$:

$$KPSS = \frac{1}{T^2} \sum_{t=1}^T \frac{S_t^2}{\hat{\sigma}_{LR}^2}$$

where $\hat{\sigma}_{LR}^2$ is a HAC long-run variance estimate.

- $H_0: y_t$ is **stationary** (note: the *reverse* of ADF).
- $H_1: y_t$ has a unit root.
- **Decision:** reject H_0 (conclude non-stationary) if KPSS exceeds its critical value / $p < 0.05$.

Why use both ADF and KPSS: ADF has notoriously low power (it often fails to reject even for stationary series). Pairing it with KPSS's opposite null gives a **confirmatory** read. We want, on the **levels**: ADF *fails to reject* **and** KPSS *rejects* → both point to non-stationary → confident $I(1)$. On the **differences** the pattern flips.

4.3 DF-GLS (Elliott–Rothenberg–Stock) — higher power

Same null/alternative as ADF, but the data are **GLS-detrended** before the ADF regression, which yields substantially **greater power** near the unit-root boundary. Include it because ADF alone is weak.

4.4 Phillips–Perron (PP) — robust to heteroskedasticity

Same H_0 (unit root) as ADF, but instead of adding lagged differences it applies a **non-parametric HAC correction** to the test statistic. Apt for FX, where volatility clustering violates the i.i.d.-error assumption.

4.5 Zivot–Andrews — break-robust unit root

- H_0 : unit root **with no structural break**.
- H_1 : stationary **around one endogenously-chosen break** in level/trend.
- The break date is chosen to be **maximally favourable to rejecting** H_0 (minimises the ADF t -stat over all candidate break points).

Why this is essential (a high-mark point): an unmodelled structural break makes a *stationary* series look like a unit root (Perron 1989). We *expect* breaks (COVID, the 2022 hiking cycle — dated in §11), so an $I(1)$ verdict from plain ADF alone would be self-contradictory. Zivot–Andrews closes that hole.

4.6 Critical-values note (remember this — it returns in §6, reversed)

Here the series are **directly observed**, so the **standard Dickey–Fuller critical values are correct**. This is *deliberately different* from §6, where we test the *estimated residual* of a cointegrating regression and must use **stricter** critical values.

4.7 What we found

- **Levels:** unit-root nulls (ADF/DF-GLS/PP/ZA) generally **not rejected**; KPSS **rejects** stationarity ($p \approx 0$) for all seven. $\rightarrow I(1)$.
- **Differences:** ADF/DF-GLS/PP all $p \approx 0.000 \rightarrow$ stationary $\rightarrow$ **confirms $I(1)$, rules out $I(2)$** .
- Two minor cross-test disagreements (MXN DF-GLS $p = 0.014$; NZD Zivot–Andrews $p = 0.019$) are expected when running five tests; the **preponderance** says $I(1)$, and the notebook flags them honestly.

5. §3 — Lag selection and the residual-whiteness gate

Johansen rests on an underlying VAR(p) in levels:

$$y_t = \nu + \sum_{i=1}^p A_i y_{t-i} + \varepsilon_t.$$

5.1 Information criteria

Choose p to minimise an information criterion, each trading fit against parameter count $k = pn^2$:

$$\text{AIC} = \ln |\hat{\Sigma}| + \frac{2k}{T}, \quad \text{BIC} = \ln |\hat{\Sigma}| + \frac{k \ln T}{T}, \quad \text{HQIC} = \ln |\hat{\Sigma}| + \frac{2k \ln \ln T}{T}.$$

The VECM lag in **differences** is $k_{\text{ar diff}} = p - 1$.

- **Our choice: AIC, floored at $p = 2$** (so $k_{\text{ar diff}} = 1$). *Why AIC, not BIC?* For cointegration the consistent criteria (BIC/HQIC) tend to **under-select** short-run dynamics, leaving residual autocorrelation that **biases the trace test**. Result: AIC = 2, BIC/HQIC = 1.

5.2 Portmanteau / Ljung–Box whiteness test (the gate)

- H_0 : VAR residuals are **white** (no autocorrelation up to lag h).
- **Statistic (multivariate, adjusted):** $Q_h = T^2 \sum_{j=1}^h (T-j)^{-1} \text{tr}(\hat{C}_j' \hat{C}_0^{-1} \hat{C}_j \hat{C}_0^{-1}) \sim \chi^2$, where $\hat{C}_j$ is the residual autocovariance at lag j .
- **Why it is a gate:** the trace test's asymptotic distribution assumes serially-uncorrelated errors, so whiteness is a **precondition**, not an afterthought.

5.3 What we found — and how to present the "failure"

The whiteness test **rejects** ($p \approx 0$). **Do not be defensive about this — own it:**

At **daily** frequency strict whiteness is essentially unattainable: the portmanteau statistic also absorbs **ARCH / volatility clustering**, which no lag length removes. Adding lags would chase conditional heteroskedasticity, not mean dynamics, and merely over-parameterise the system. We therefore keep the AIC order and obtain

valid rank inference from the **heteroskedasticity-robust wild bootstrap in §5**, rather than the asymptotic test alone.

This turns a nuisance into a rigour point — it *motivates* the bootstrap.

6. §4 — Deterministic specification (which of Johansen's 5 cases)

Writing the VECM with deterministics split between the levels and the cointegrating relation:

$$\Delta y_t = \alpha(\beta' y_{t-1} + \rho_0 + \rho_1 t) + \gamma_0 + \gamma_1 t + \sum_{i=1}^k \Gamma_i \Delta y_{t-i} + \varepsilon_t.$$

Case	Levels	Coint. relation	Meaning
1	–	–	no deterministics
2	–	constant ρ_0	equilibrium has non-zero mean; no trend in levels
3	constant γ_0	constant	linear trend allowed in levels ('co')
4	constant	+ trend ρ_1	trend-stationary equilibrium
5	trend	+ trend	quadratic trends

The choice changes the critical values, so it is substantive.

- **Our choice — Case 3 (unrestricted constant, deterministic='co')**. §1 shows EM currencies (ZAR, MXN, NOK) **drift persistently** vs USD. We confirm with a regression $y_t = a + bt$: HAC t -stats on the slope are **huge** (e.g. NZD $t = -27$, NOK/ZAR $t \approx -15$) → a real deterministic trend the model must accommodate → unrestricted constant. Case 2 is run as a robustness check.

7. §5 — The Johansen cointegration test (the core inference)

7.1 From VAR to VECM

Re-parameterise the VAR(p) as the **Vector Error-Correction Model**:

$$\Delta y_t = \Pi y_{t-1} + \sum_{i=1}^k \Gamma_i \Delta y_{t-i} + (\text{determ.}) + \varepsilon_t, \quad \Pi = \alpha \beta'.$$

- Π is the **long-run impact matrix**. Its **rank** is the cointegration rank: $r = \text{rank}(\Pi)$.
- If $r = 0$: no cointegration ($\Pi = 0$, a VAR in differences). If $r = n$: the system is stationary in levels. If $0 < r < n$: **r cointegrating relations**, with
- $\beta (n \times r)$: the **cointegrating vectors** (the equilibria),
- $\alpha (n \times r)$: the **adjustment / loading matrix** (speeds of return to equilibrium).

7.2 Reduced-rank regression & eigenvalues

Johansen's MLE concentrates out the short-run terms and solves a generalised eigenvalue problem on the residual moment matrices S_{ij} (from regressing Δy_t and y_{t-1} on the lagged differences):

$$|\lambda S_{11} - S_{10}S_{00}^{-1}S_{01}| = 0 \Rightarrow \hat{\lambda}_1 > \hat{\lambda}_2 > \dots > \hat{\lambda}_n.$$

Each $\hat{\lambda}_i$ is a **squared canonical correlation** between levels and differences; a large $\hat{\lambda}_i$ signals a stationary (mean-reverting) direction.

7.3 The two rank tests

Trace test — tests rank $\leq r$ against rank $> r$:

$$\text{LR}_{\text{trace}}(r) = -T \sum_{i=r+1}^n \ln(1 - \hat{\lambda}_i).$$

- H_0 : at most r cointegrating relations. H_1 : more than r .
- Procedure: start at $r = 0$; if rejected, try $r = 1, \dots$ stop at the first **non-rejection**.

Maximum-eigenvalue test — tests rank $= r$ against rank $= r + 1$:

$$\text{LR}_{\text{max}}(r) = -T \ln(1 - \hat{\lambda}_{r+1}).$$

- Sharper (isolates one eigenvalue); can disagree with trace. **We report both and reconcile them.**

Both have **non-standard** asymptotic distributions (functionals of Brownian motion) that depend on $n - r$ and the deterministic case; critical values come from MacKinnon–Haug–Michelis tables (bundled in `statsmodels`).

7.4 Two robustness layers (these are what elevate the section)

(a) Reinsel–Ahn finite-sample correction. The trace test **over-rejects** in finite samples. Replace T by the effective sample $T - nk$:

$$\text{LR}_{\text{trace}}^*(r) = -(T - nk) \sum_{i>r} \ln(1 - \hat{\lambda}_i).$$

(b) Wild bootstrap (Cavaliere–Rahbek–Taylor). Asymptotic critical values assume homoskedastic Gaussian errors; FX has heavy volatility clustering that inflates the trace statistic. Algorithm for testing $H_0 : r = 0: 1$. Estimate the restricted model under H_0 (a VAR in differences); keep residuals $\hat{\varepsilon}_t$. 2. Generate **Rademacher wild draws** $w_t \in \{-1, +1\}$ and build $\varepsilon_t^* = w_t \hat{\varepsilon}_t$ — this **preserves the conditional heteroskedasticity** pattern. 3. Recursively rebuild a bootstrap series y_t^* and recompute $\text{LR}_{\text{trace}}(0)$. 4. Repeat B times; bootstrap p -value
$$= \frac{\#\{\text{LR}^* \geq \text{LR}^{\text{obs}}\} + 1}{B + 1}.$$

(The notebook's from-scratch reduced-rank routine is validated to match `statsmodels.coint_johansen` to the decimal, and supplies the S_{ij} matrices reused in §8.)

7.5 What we found — the headline

Quantity	Value
Trace rank @95%	0
Finite-sample-corrected trace rank	0
Max-eigenvalue rank @95%	0
Observed $LR_{\text{trace}}(0)$	115.58
Asymptotic 95% crit. value	125.62
Bootstrap 95% crit. value	135.62
Bootstrap p-value	0.345

How to present this beautifully: the bootstrap critical value (135.6) sits **above** the asymptotic one (125.6) — *exactly* the direction theory predicts when errors are heteroskedastic (the asymptotic test over-rejects). All four routes agree: **no cointegration in the full 2016–26 sample**. This is not a failure — it is the FX-instability thesis, confirmed by convergent, robust tests.

8. §6 — Cross-validating cointegration (a second method)

Relying on one test is fragile; we confirm with **residual-based** tests and re-estimate β two more ways.

8.1 Engle–Granger & Phillips–Ouliaris

Regress one currency on the others, $y_{1t} = \theta' y_{-1,t} + u_t$, and test the residual $\hat{u}_t$ for a unit root. - H_0 : $\hat{u}_t$ has a unit root $\rightarrow$ **no cointegration**. H_1 : $\hat{u}_t$ stationary $\rightarrow$ cointegration. - **Engle–Granger** uses an ADF on $\hat{u}_t$; **Phillips–Ouliaris** uses a PP-type (HAC) statistic on $\hat{u}_t$.

The critical-values point (the §2 caveat, now reversed): because $\hat{u}_t$ is the residual of an **estimated** regression whose coefficients were chosen to *minimise its variance*, $\hat{u}_t$ is biased toward looking stationary. So we **must use the stricter, N -dependent Engle–Granger / Phillips–Ouliaris critical values** (more regressors $\Rightarrow$ more stringent). `arch` applies these automatically; feeding $\hat{u}_t$ into a plain ADF would silently use too-lenient values and **over-detect** cointegration. Make this point explicitly — it shows you understand *why* there are two sets of critical values.

8.2 DOLS (Stock–Watson) and the Johansen β

Re-estimate the long-run vector with valid standard errors (DOLS adds leads and lags of Δy to correct for endogeneity). **Convergent β across Johansen / DOLS is strong evidence; divergence is itself a finding.**

8.3 What we found

Engle–Granger $p = 0.61$, Phillips–Ouliaris $p = 0.69 \rightarrow$ **no cointegration**, independently confirming Johansen. (Long-run vectors are shown for comparison but flagged as economically meaningful only if $r \geq 1$.)

9. §7 — VECM estimation & diagnostics

Fit the VECM at the chosen rank:

$$\Delta y_t = \underbrace{\alpha}_{\text{adjustment}} \underbrace{\beta'}_{\text{equilibria}} y_{t-1} + \sum_{i=1}^k \Gamma_i \Delta y_{t-i} + (\text{determ.}) + \varepsilon_t.$$

- β — long-run equilibrium combinations.
- α — how fast each currency is pulled back to equilibrium (the "which to hold" engine, §9).

Three checks — "a model we have not checked is a model we cannot trust": 1. **Stationarity of the error-correction term** $\hat{\beta}' y_t$ (ADF/KPSS) — directly verifies the cointegrating combination is $I(0)$, closing the loop.

Flagged as indicative only, because β is estimated (the §6 caveat); formal evidence is the §6 properly-valued tests. 2.

Companion-matrix stability — the VAR companion form must have exactly $n - r$ unit roots and all other eigenvalue moduli < 1 . 3. **Residual whiteness / normality** on the fitted VECM.

Presenter note: since the full-sample rank is 0, the notebook sets rank = 1 **only to demonstrate the estimation/diagnostics machinery**; the honest rank verdict (0) stands. Say this plainly.

10. §8 — Testing the common-dollar-factor trap (an LR test)

The risk: because every series is vs USD, a "cointegrating relation" might just be **shared dollar exposure**. The purest dollar relation is the **equal-weight direction** $b = (1, \dots, 1)'$. We test whether the estimated cointegrating space is statistically consistent with that direction.

For rank $r = 1$, restricting β to a known vector b gives the restricted squared canonical correlation and a **likelihood-ratio statistic**:

$$\lambda^* = \frac{b' S_{10} S_{00}^{-1} S_{01} b}{b' S_{11} b}, \quad \text{LR} = T \ln \frac{1 - \lambda^*}{1 - \hat{\lambda}_1} \sim \chi_{n-1}^2.$$

- H_0 : the cointegrating vector **equals** the dollar direction $b = (1, \dots, 1)'$ → the relation is a pure dollar factor (untradeable as relative value).
- H_1 : $\beta \neq b$ → a genuine cross-currency relationship.
- **Decision:** small $p \Rightarrow$ **reject** $\Rightarrow$ *not* a dollar factor.

What we found: $\hat{\lambda}_1 = 0.0146$, $\lambda^* = 0.0047$, $\text{LR} = 26.07$, $\chi_6^2 p = 0.0002 \rightarrow$ **reject** H_0 : the dominant direction is **statistically distinguishable from a pure dollar factor**. (We tested the worry rather than merely asserting it — an A-vs-B+ move.)

11. §9 — Structural interpretation: *which currencies to hold*

11.1 Weak exogeneity (anchors vs adjusters)

A currency whose loading $\alpha_j \approx 0$ does **not** error-correct — it **pushes** the system but is not pulled back: a long-run **anchor** / **driver**. Large, significant $\alpha_j \Rightarrow$ an **adjuster** that reverts toward the basket. - $H_0: \alpha_j = 0$ (currency j is **weakly exogenous**). Formally an LR test; we report the asymptotically-equivalent t -tests on each α_j . - **Decision:** reject ($p < 0.05$) $\Rightarrow$ adjuster; fail to reject $\Rightarrow$ anchor.

What we found (the holding answer):

Currency	α	t	p	Role
AUD	-0.0013	-2.07	0.038	Adjuster
NZD	-0.0013	-2.08	0.038	Adjuster
CAD	-0.0006	-1.34	0.179	Anchor
NOK	-0.0026	-3.55	0.000	Adjuster
SEK	-0.0023	-3.60	0.000	Adjuster
ZAR	+0.0018	1.92	0.055	Anchor
MXN	-0.0001	-0.18	0.858	Anchor

$\rightarrow$ **CAD, ZAR, MXN are anchors** (natural core holdings the basket is pinned to); **AUD/NZD/NOK/SEK are adjusters** (where the mean-reversion trade lives).

11.2 Half-life of mean reversion

Model the error-correction term as an AR(1), $z_t = c + \phi z_{t-1} + e_t$. The **half-life** of a shock:

$$H = \frac{\ln(0.5)}{\ln \phi} \quad (\text{trading days}).$$

What we found: $\phi = 0.9789 \Rightarrow H = 32.4$ days. We then **set the walk-forward holding window $M = 32$ from this half-life** rather than picking it arbitrarily — a derived, not assumed, parameter.

(Identification note: with $r > 1$, β is identified only up to rotation and needs restrictions to interpret; with $r = 1$ this is moot.)

12. §10 — From equilibrium error to an allocation

The tradeable object is the **standardised error-correction term**:

$$s_t = \hat{\beta}' y_t, \quad z_t = \frac{s_t - \mu}{\sigma},$$

with μ, σ estimated **without look-ahead** (train/expanding only). When z_t is stretched, the rich legs are overvalued vs the cheap legs, so we **lean against the deviation**:

$$w_t = -z_t \hat{\beta} / \|z_t \hat{\beta}\|_1.$$

Hold the cheap legs, fund the rich legs, gross-normalised to 1 — the explicit "which to hold, and how much, today" rule.

13. §11 — Walk-forward out-of-sample (the headline section)

Mechanics — say this slowly, it's the part people misunderstand:

We are **not** forecasting exchange rates or "predicting cointegration." At each rebalance point t_0 : 1. Take the **trailing** window $[t_0 - N, t_0)$, $N = 504$ days (≈ 2 yr) — past data only. 2. Run Johansen. **If $r \geq 1$:** fit the VECM, **freeze** β, μ, σ . **If $r = 0$:** **hold cash** for the next block (encode the instability, don't force a trade). 3. For the next $M = 32$ days, trade the frozen signal; the position formed at close t earns the $t \rightarrow t+1$ return. 4. Advance by M , **refit**. Stitch all out-of-sample blocks into one continuous, **fully-causal** equity curve.

- **Rolling (fixed- N)**, not expanding — we *want* the model to forget pre-regime data; that adaptivity is the right choice for an unstable relationship.
- **Leakage discipline:** every quantity used at t uses only data $\leq t$; an explicit assertion checks train indices precede the traded returns.

What we found: - Leakage self-check: **PASS**. - OOS span 2018-05 $\rightarrow$ 2026-06 (2,102 days); **in-market only 15% of days** (cash 85%); **15% of refit windows have rank ≥ 1** . - β drifts and the rank flickers between 0 and ≥ 1 — *that drift is the result: FX cointegration is regime-dependent, not a fixed structural law.*

§11b — descriptive stability (firewalled from the signal): - **Recursive trace statistic** (expanding window, scaled by its 95% CV): wanders above/below 1, showing the relationship's existence is not constant. - **Bai–Perron / PELT break detection** on the ECT (via `ruptures`) dates the regime shifts. *Crucially, full-sample break dates are never fed back into the signal* — that would be look-ahead.

14. §12 — Out-of-sample statistical evaluation (graded core)

The grade is on **statistics**, so we evaluate statistically, not just by the equity curve. The benchmark every FX model must clear is the **random walk** (Meese–Rogoff 1983: exchange-rate changes are close to unforecastable).

14.1 Predictive regression (the mean-reversion hypothesis, formally tested)

$$\Delta s_{t+1} = a + b z_t + u_{t+1}, \quad \text{Newey–West (HAC) standard errors.}$$

- $H_0: b = 0$ (the equilibrium error has no predictive content). $H_1: b < 0$ (deviations are partly reversed — error correction).
- **HAC is mandatory:** autocorrelation + volatility clustering would otherwise overstate the t -stat. Newey–West uses $\hat{\sigma}_{LR}^2 = \hat{\gamma}_0 + 2 \sum_{j=1}^L (1 - \frac{j}{L+1}) \hat{\gamma}_j$.

What we found: $b = -0.00398$, HAC $t = -3.47$, $p = 0.001 \rightarrow$ **statistically significant mean reversion**.

14.2 Diebold–Mariano test vs a random walk

Compare one-step forecasts of s_t from the error-correction model against the random-walk forecast $\hat{s}_{t+1} = s_t$. With loss differential $d_t = e_t^{\text{RW}^2} - e_t^{\text{VECM}^2}$:

$$DM = \frac{\bar{d}}{\sqrt{\widehat{\text{Var}}(\bar{d})}} \xrightarrow{d} \mathcal{N}(0, 1), \quad \widehat{\text{Var}}(\bar{d}) \text{ from a HAC long-run variance.}$$

- H_0 : equal forecast accuracy. H_1 : the model forecasts better ($DM > 0$).
- **What we found:** $DM = 1.34, p = 0.18 \rightarrow$ **cannot reject equal accuracy: the model does not beat the random walk.**

The sophisticated punchline to land: there is *statistically significant local mean reversion* in the equilibrium error (predictive regression), yet that **does not translate into beating a random walk** (Diebold–Mariano). This is precisely the Meese–Rogoff nuance — predictability $\neq$ forecast superiority — and reporting it honestly is exactly what the course rewards.

15. §13 — Comparative benchmarks (the demonstration)

Over the **identical OOS window**, compare the strategy against naïve alternatives: each single-currency buy-and-hold, **USD cash** (the flat "do nothing" line), and an equal-weight basket.

What we found (price-only log returns):

Strategy	Total ret	Ann. vol	Sharpe	Max DD
Cointegration (walk-fwd)	+0.030	0.011	+0.34	−0.02
Hold AUD	−0.070	0.100	−0.08	−0.29
Hold NZD	−0.170	0.098	−0.21	−0.30
Hold CAD	−0.084	0.064	−0.16	−0.19
Hold NOK	−0.156	0.120	−0.16	−0.38
Hold SEK	−0.084	0.103	−0.10	−0.33
Hold ZAR	−0.276	0.141	−0.23	−0.47
Hold MXN	+0.122	0.120	+0.12	−0.32
USD cash	0.000	0.000	—	0.00
Equal-weight basket	−0.103	0.086	−0.14	−0.25

Two honesty caveats — state both: 1. **Risk profiles differ.** The strategy is **long/short, dollar-neutral** (it strips out USD direction); the buy-and-holds carry full directional USD exposure. Not apples-to-apples — that's the point: it isolates relative value, and by sitting in cash 85% of the time it sidestepped the EM drawdowns. 2. **Carry is omitted.** Price-only P&L misses rate differentials. Real holders of high-yielders (ZAR, MXN) earn substantial **carry** $\rightarrow$ the single-currency lines are **understated** (e.g. real MXN returns would be materially higher). We do **not** claim "the strategy beats holding MXN" without that asterisk.

16. Headline results — one slide

Question	Test(s)	Verdict
Integration order	ADF, KPSS, DF-GLS, PP, Zivot–Andrews	all seven $I(1)$
Lag / specification	AIC + portmanteau; HAC drift test	$p = 2$; Case 3 (drift)
Cointegration (full sample)	Johansen trace & max-eig, Reinsel–Ahn, wild bootstrap , EG, PO	rank 0 — no stable cointegration
Is it just the dollar?	β -restriction LR (χ^2_6)	rejected — not a pure dollar factor
Which to hold?	weak-exogeneity t -tests	anchors CAD/ZAR/MXN ; adjusters AUD/NZD/NOK/SEK
Reversion speed	AR(1) half-life	32 days
Is it stable over time?	rolling Johansen, recursive trace, Bai–Perron	regime-dependent — coint. in only 15% of windows
Local predictability?	predictive regression (HAC)	yes , $b < 0$, $t = -3.5$
Beats a random walk?	Diebold–Mariano	no ($p = 0.18$) — Meese–Rogoff holds

The story in one line: a fully-diagnosed pipeline showing that cointegration in this FX basket is real but fragile and regime-dependent, with mean reversion that is statistically significant yet not forecast-superior to a random walk.

17. Anticipated questions (and crisp answers)

- **"Isn't using the whole sample look-ahead bias?"** Only for *predictive* claims. §2–§9 are *descriptive* in-sample inference (no forecast); all *predictive* work (§11–§13) is strictly causal with a leakage assertion.
- **"Your whiteness test failed — isn't the model invalid?"** At daily frequency strict whiteness is unattainable (ARCH dominates the portmanteau). That's *why* we use a heteroskedasticity-robust **wild bootstrap** for rank inference instead of relying on asymptotics.
- **"Why ADF and KPSS?"** Opposite nulls → a confirmatory read that guards against ADF's low power.
- **"Why is the cointegration ADF different from a normal ADF?"** The residual of an estimated cointegrating regression is biased toward stationarity (coefficients chosen to minimise its variance), so it needs **stricter**, N -**dependent** Engle–Granger/Phillips–Ouliaris critical values — *not* standard DF values.
- **"Rank is 0 — isn't the project a failure?"** No. The negative result is *established by convergent robust tests*, and the walk-forward shows the relationship genuinely flickers in and out. Honest characterisation of an unstable relationship is the deliverable.
- **"Why trace and max-eigenvalue?"** They test different hypotheses ($\leq r$ vs $= r$) and can disagree; reporting both and reconciling is the rigorous standard.
- **"Why rolling, not expanding, windows?"** We *want* the model to forget pre-regime data; adaptivity suits an unstable relationship.
- **"How does VECM answer which to hold?"** Two ways: structurally via weak exogeneity (anchors vs adjusters) and dynamically via the error-correction signal (lean against z_t).

18. Mini-glossary

- $I(d)$ — integrated of order d ; needs d differences to be stationary.
 - **Cointegration** — a stationary linear combination $\beta'y_t$ of non-stationary series; a long-run equilibrium.
 - **Rank** r — number of independent cointegrating relations = $\text{rank}(\Pi)$.
 - β — cointegrating vectors (equilibria). α — adjustment speeds (loadings).
 - **Error-correction term (ECT)** — $z_t = \beta'y_t$, the disequilibrium; mean-reverting if cointegrated.
 - **Weak exogeneity** — $\alpha_j = 0$: currency j drives but does not adjust (an anchor).
 - **Half-life** — $\ln 0.5 / \ln \phi$; time for a deviation to halve.
 - **HAC / Newey–West** — heteroskedasticity- and autocorrelation-consistent standard errors.
 - **Wild bootstrap** — resampling with random sign flips that preserves volatility clustering.
 - **Meese–Rogoff** — the result that structural FX models struggle to beat a random walk OOS.
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Reproduce everything: `uv run --with nbformat python src/build_notebook.py` regenerates the notebook;
run it on the project `.venv` from the `multivariate_project/` directory.